ELCT 363: Introduction to Microelectronics **Homework #5**
University of South Carolina/ Spring 2020

Due by:

Student's Name: ____

Problem : (3+3 = 6 points)

In LED what are the three different efficiency components, define reach clearly. Calculate the internal quantum efficiency of a LED which has overall efficiency of 30% with injection and extraction efficiencies of 70% and 50%.

Solution:
Look at the definitions in the class notes:
Injection Efficiency >>>> η_I
Internal Quantum Efficiency or
Quantum Efficiency or Recombination Efficiency >>>> η_r
Extraction Efficiency >>>> η_E

Overall Efficiency, $\boldsymbol{\eta} = \eta_I \,.\, \eta_r \,.\, \eta_E$

Internal Quantum Efficiency, $\eta_I = \boldsymbol{\eta}/\eta_r \,.\, \eta_E$ = 0.3/(0.7x0.5) = 0.86 Or 86%

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Problem : (3 point each = 6 points)

What are the differences between LEDs and Laser diodes (LD)

Solution:

LED

- Emission in random direction
- Transition is spontaneous
- No need of cavity for increase in the light

LD

- Emission is directional
- Transition is stimulated by photons
- Need a specific cavity length to amplify the light

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Problem : (3point each = 6 points)

What would be the ideal wavelength which GaN and GaAs Light Emitting Diodes (LEDs) can produce?

Solution:

For GaN:

Si Bandgap, Eg = 3.4 eV converting it to the wavelength $\lambda = 1.24/Eg = 0.3645\ \mu m$

Thus, a GaN based LED will ideally produce a wavelength of 0.3647 µm or 364.7 nm

For GaAs:

GaAs Bandgap, Eg = 1.42eV converting it to the wavelength $\lambda = 1.24/Eg = 0.87\ \mu m$

Thus, a Si based LED will ideally produce a wavelength of 0.87 µm.

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Problem : (1+1+1+1+1+1 = 6 points)

What are differences between bipolar junction transistors and field effect transistors, describe/sketch HEFET, MOSFET, MISFET, MOSHFET, MISHFET.

The are sketches were described in the class and are in the notes.

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Problem : (3+3 = 6 points)

Define High electron mobility transistor and describe how 2DEG can make this transistor faster than other transistors. Write it in detail.

Well discussed in the class and described in the class notes, the 2DEG and give reason of less scattering in due to absence of doping.